THE UNIVERSITY OF TEXAS AT DALLAS

ENTP 4332 - Fall 2026
AI in Action: Building Products, Ventures, and Competitive Advantage
Syllabus Version Date: September 25, 2026

Instructor

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Course Code

ENTP 4332. This class is cross-listed with ITSS 4332.

Class Schedule for Fall 2026

Section 001: Tuesday, 1 - 3:45PM, Room: SOM 22.110

Office Hours

11am-12pm on Tuesdays in JSOM 4.222

Students are welcome to visit office hours *without an appointment* to discuss course details (e.g., logistics, material, assignments, exams). If the above time does not work for you, or if you have something specific to discuss, **please email me to schedule an appointment**.

Course Description

This course focuses on building new products and ventures with the help of artificial intelligence (AI). There are two parts to this course. First, students will learn how to use AI to conduct market research, formulate strategy, assess competition, and identify customer needs. Second, students will use AI to build new products. Working in teams, they will create AI-assisted strategic analyses, functional prototypes, go-to-market strategies, and investor-ready pitch decks. Throughout the course, students will also learn how AI is reshaping the nature of competition, viable business models, and the structure of organizations, and get to hear first-hand accounts from industry executives and startup founders. The course combines hands-on practice with strategic thinking, helping students leverage cutting-edge AI tools, while also understanding their broader implications for strategy and society.

Course Objectives

By the end of this course and the completion of all deliverables, students will be able to take an idea to a working product with the help of AI tools. This includes the following four course objectives:

1. First, students will be able to build, test, and deploy a working digital product by directing AI coding tools. Specifically, students will write a product requirements document; build and deploy a publicly accessible web application under version control; add a data layer and at least one additional

component to that application; identify and fix problems in AI-generated products; and improve the product's design and interface based on usability principles and feedback from real users.

2. Second, students will be able to use AI to research and evaluate the market opportunity linked to their product. This includes defining the market problem their product solves and the argument for current timing of entry; estimating the size of the target market with reasonable evidence; designing and conducting customer interviews and summarizing what they learned; identifying and comparing direct competitors; and explaining what would make their product difficult to copy. Throughout, students will practice checking AI-generated claims against real sources rather than accepting them at face value.

3. Third, students will be able to pitch the product and business plan in a live a presentation, presenting the customer problem and the value delivered rather than the technology, and answering questions about their assumptions.

4. Fourth, students will be able to describe how AI is changing work, industries, and entrepreneurship, including its ethical and workforce implications, and connect those changes to their own product.

Required Course Materials

Students must purchase a Claude Pro subscription ($20/month + tax). If UT-Dallas changes institutional access to Claude by the start of the class, there might be the possibility to use it instead of purchasing a subscription. Other AI tools (e.g., Open AI's Codex or Google Antigravity) may be used upon discussion with the instructor. All other tools that will be used in the course have sufficient free tiers. Instructions for access of each are available on the course website, and will also be provided in class.

Other than the above, there are no other materials to be purchased. Required and optional course readings (noted in the course schedule below) are available online for free, and where possible, in Canvas.

Course Evaluation

Course grades will be based on the following four components:

1. ***Class Participation (Individual Assignment; Weightage: 20%)***
   Active engagement in the course plays an integral role in the learning process for students and their peers, and can be achieved through participation in class and in assignment groups. Class participation will be evaluated as a function of the following:

   - Contribution to in-class discussions: A critical component of the course experience will be rigorous and spirited discussion. I expect all students to be well prepared for each class, having studied the required pre-class material. Students who make consistent, meaningful contributions to class discussions will receive better class participation grades. The quality of comments is evaluated based on the clarity and coherence of ideas and arguments, and the ability to support claims with quantitative or qualitative evidence from the course materials (e.g., slides, readings). Finally, engagement by colleagues during student group presentations is also acknowledged.

     - Name Tents: I will provide students with name tents (i.e., a prominent sign to write full name on) on the first day of class. Students must bring these name tents to every subsequent session. This is critical for accurate receipt of participation points.

2. ***Attendance (Individual Assignment; Weightage: 5%)***
   Attendance is a student's obligation, and comprises 5% of the student's overall course grade. No right or privilege exists that permits you to be absent from any class meetings except for excused absences for authorized University activities or Religious Holy Days.

- Two unexcused absences per semester are allowed for which no points will be deducted. For these absences, students do not need to inform me.

- Late attendance to class: Arriving to class past the start time will be marked as late. Excessive late attendance to class will result in a reduction of participation points.

3. ***First App and Build Memo (Individual Assignment; Weightage 5%)***
   In this assignment, students will build and deploy a small working web application using AI. There are two parts two this assignment. First, students will submit a live URL for the web application, and a Github repository link. Second, students will submit one-page single-spaced memo that explains what was built, and why it was built it (i.e., what market problem did it solve, and how will customers use the application).

4. ***Individual Build Log (Individual Assignment; Weightage 15%)***
   Starting Session 4 onwards, each week students must maintain a build log. It should be one page long, and discuss what you built that week, what unique things you tried, what broke along the way, how you fixed it, and what you learned. At the end of the semester, students will submit their four most interesting logs in a combined document. In the final submission, students must also describe a case where AI was confidently wrong, how the student caught it, and learnings from these incidents. I will grade the final combined build log as: "Poor", "Average", "Good", or "Excellent", which will correlate to particular scores for this assignment.

5. ***Product Proposal (Team Assignment; Weightage: 5%)***
   The Product Proposal is a team-written 2-page project proposal outlining the group's two potential ideas for a new product. This should briefly cover why the group is interested in this idea, what problem it will solve, the novelty, value, and uniqueness of the proposed solution, and why AI can help build it. More details will be available on Canvas.

6. ***Progress Week Meeting (Team Assignment; Weightage: 5%)***
   In Session 7, each student group will meet with me during class to discuss the progress they have made on their discovery report. I will grade each group's progress as: "Poor Progress", "Marginal Progress", "Good Progress", or "Excellent Progress", which will correlate to particular scores for this assignment.

7. ***Discovery Report (Team Assignment; Weightage: 15%)***
   The Discovery Report is a team-written summary of the research each group conducts across four sessions: *Finding Market Problems*, *Understanding Customers*, and *Assessing Competition & Moats*. In the assignment, teams will explicitly identify and define the market problem they are solving and explain why now is the right time to target the market. In addition, students will size the target market, present what they learned from customer research, analyze their competitors, argue for the defensibility of their product's proposed moat, and lay out a business model with planned unit economics. Students will use AI throughout to conduct this research but must verify and validate its output. In addition, AI is not permitted for the writing of the report. The report is due as a single document, with a total of eight double-spaced pages maximum plus appendices and references.

8. ***MVP Demo (Team Assignment; Weightage: 10%)***
   Students must complete an six-minute live, working demonstration of their product's core workflow in front of the class. The following must also be submitted via Canvas: live URL, link to Github repository, and relevant markdown files. This assignment will be graded on whether the product runs, whether evidence presented in the Discovery Report visibly shaped the product and its features, the quality of your testing and validation, and responses to questions during the presentation.

9. ***Final Venture Package and Pitch Presentation (Team Assignment; Weightage: 20%)***
   At the end of the semester, each team will deliver an eight-minute pitch presentation, followed by five

minutes of Q&A. Teams will then submit a final package containing the final pitch deck with an appendix of supporting evidence, a live product URL, a GitHub repository, and a two-page double-spaced AI-usage report documenting the most impactful uses of AI over the semester, such as pivotal prompts and agent workflows. The package also includes the Discovery Report, but now revised and reframed as a G*o-to-Market Strategy* document. This is an update rather than a rewrite. Therefore, every substantive change must be visibly annotated using highlighting or a short change log so it clearly evident to see what shifted and why. Finally, a central addition to this deliverable is the incorporation of real customer evidence. Finally, a central addition to this deliverable is the incorporation of real customer evidence. Each student must document a minimum of four real interviews (even if working in a team), and learnings from each. More details will be provided in class and on Canvas.

10. ***Peer Evaluation***
    A confidential end-of-term evaluation adjusts individual grades on team components by a multiplier of 0.95 to 1.05. Individual assignments are not subject to multipliers.

Final Course Grades

Final grades will be assigned based on the following scale: 93-100% (A), 90-92.99 (A-), 87-89.99 (B+), 83-86.99 (B), 80-82.99 (B-), 77-79.99 (C+), 73-76.99 (C), 70-72.99 (C-), and so forth. I reserve the right to make changes to this distribution based on overall class performance at the end of the semester. No exceptions or rounding up of grades will be provided.

Other Class Policies

1. ***The Use of Generative AI Tools in the Classroom & Assignments.***
   This class requires the use of generative AI. However:

   - **Every factual claim in submitted assignments needs to be manually verified.** AI generates confident, plausible, and often fabricated citations and claims. Whenever you submit an assignment, you are responsible for verifying the accuracy of the claims.

   - **Your AI-usage documentation must be truthful.** The build log and the prompt book are graded on insight, and not on how little AI you used. There is no advantage to understating usage; rather, more thoughtful and creative use of AI is encouraged.

   - **No AI-slop.** By now, you have likely realized that writing generated by AI is often difficult to read or convoluted in its logic and evidence. For written assignments, I encourage students to plan to write without AI, and only use AI for revising and iterating on arguments and flow. This will significantly enhance the quality of writing.

   - **UTD policy.** Please review the university policy on **Generative AI use in academic work**.

2. ***The Use of Smart Devices in the Classroom***
   Electronic devices can disrupt learning for everyone in the classroom, particularly in a discussion-based course where full engagement matters. For this reason, no devices (including phones, laptops, or tablets) are allowed in the classroom, except during lab sessions (i.e., the second half of class), and personal emergencies as defined next. A device may be used briefly for a personal emergency, such as a health, family, or childcare situation. Where possible, please notify the instructor before class if you expect to need this. Students with an approved accommodation through the AccessAbility Resource Center (ARC) that requires device use, such as for note-taking, should contact the instructor directly to arrange this. Printed notes or other hand-written notes are permitted. Slides will be posted to Canvas within 24 hours of class. If a device is used outside of these exceptions, the student will first receive a reminder. Repeated instances will result in a large reductions of participation points.

3. ***University Course Policies***
   Information contained in the following link constitutes additional University's policies and procedures. Please go to UT Dallas Syllabus Policies webpage for these policies (**https://go.utdallas.edu/syllabus-policies**).

4. ***Honor Code***
   As members of the UT Dallas community, all students are expected to uphold the Comet Creed: *"As a Comet, I pledge honesty, integrity, and service in all that I do."* Severe implications will exist for plagiarism in any capacity, unethical academic behavior, or violations of UT Dallas' official **student code of conduct**. In the event of a violation of the honor code, I will follow UT Dallas' official reporting process for investigation.

5. ***Accommodations for Students with Disabilities***
   The University of Texas at Dallas is committed to providing reasonable accommodations for all persons with disabilities. The syllabus is available in alternate formats upon request. If you are seeking classroom accommodations under the Americans with Disabilities Act (2008), you are required to register with the AccessAbility Resource Center (ARC), located in the Administration Building, Suite 2.224. Their phone number is 972-883-2098, email: **studentaccess@utdallas.edu** and the website is **https://accessability.utdallas.edu/**. To receive academic accommodations for this class, please register and request services by completing the Request for Services form with the proper documentation and meet with the Director of ARC at the beginning of the semester.

6. **Syllabus**
   I reserve the right to make any changes to the syllabus, including readings, projects due dates, test dates, and course policies. These changes will be announced as early as possible.

Course Schedule & Weekly Task List

Session	Date	Topic
		Module 1: Foundations of AI-Enabled Product Development
1	8/25	**Introduction** *Pre-Class Task List:* 1. Read: The decline of travel agents (Ernie Tedeschi, Stripe Economics, 2026) — *optional, but highly recommended* *Topics Covered* • Current state of AI in 2026 (e..g, use cases, economic & societal impact, entrepreneurship, labor markets) • Why AI is helpful for new product development • Course grading & expectations
2	9/1	**Installing & Understanding Tools - Part I** *Pre-Class Task List:* 1. Read Course Handbook Parts 1 to 3 2. Purchase Claude Code subscription 3. Watch Software Is Changing (Again) (Andrej Karpathy, 2025) — *optional* *Topics Covered* • English as a programming language • Conceptual frameworks for building with AI • Unpacking AI-jargon (e.g., LLMs, prompts, agents, skills, human-in-the-loop) *In-Class Lab Tasks:* • Install Claude Code, explore key elements • Build first working web application
3	9/8	**Installing & Understanding Tools - Part II** *Pre-Class Task List:* 1. Submit group project team composition 2. Read Course Handbook Parts 4 to 6 *Topics Covered* • Understanding AI agents, and their use cases • Verification as the differentiator • Writing a strong product requirements documents (PRD) *In-Class Lab Tasks:* • Install git, and create first repository • Add 1-2 features to first web application

Session	Date	Topic
		Module 2: Building with Strategic Intent
4	9/15	**Finding Market Problems** *Pre-Class Task List:* 1. Submit "First App and Build Memo" assignment on Canvas 2. Prepare three potential ideas for market problems you would like to solve. For inspiration, skim through the following resources: **AI Market Clarity** (Elad Gil, 2025), **a16z's Where Enterprises Are Actually Adopting AI** (Kimberly Tan, 2026), and Requests for Startups from **Y Combinator**, **Bessemer**, **Greylock**, and **Sequioa** — *optional, but highly useful!* 3. Read Course Handbook Parts 7 and 8 *Topics Covered* • Complements, substitutes, and Jevons paradox • Vertical industry applications of AI • Assessing viability of market opportunities *In-Class Lab Tasks:* • Run an AI agent workflow to conduct opportunity evaluations on 3 candidate problem areas • Build a landing page for three candidate problem areas
5	9/22	**Understanding Customers** *Pre-Class Task List:* 1. Read Course Handbook Part 9 2. Read **Customer Interviews: How to Recruit, What to Ask, and How to Synthesize What You Learn** (Teresa Torres, 2022) — *optional* *Topics Covered* • The importance of customer discovery and validation • Using AI for customer research and assessing qualitative customer data *In-Class Lab Tasks:* • Design interview questions and structure; conduct practice interviews • Write a product requirement document (PRD) for your chosen idea, and iterate on it with AI
6	9/29	**Pricing & Business Models** *Pre-Class Task List:* 1. Submit "Product Proposal" assignment *Session Guest Workshop:* • **Garvit Patel** (Applied Scientist at Uber) *Topics Covered* • Understanding pricing strategies • Managing pricing with AI-inference *In-Class Lab Tasks:* • Draft your business model choice and pricing hypothesis

Session	Date	Topic
	10/6	**Progress Week** *Pre-Class Task List:* 1. Prepare rough plan of action for Discovery Report *In-Class Lab Tasks:* • Individual group meetings (15-20 mins each) to go over each component of your discovery report
	10/13	***Break; No In-Person Class***
8	10/20	**Mapping Product Infrastructure: From Skeleton to Real Application** *Pre-Class Task List:* 1. Read Course Handbook Part 11 *Topics Covered* • Architecture of a software product (non-technical overview) • Build vs. buy at the component-level • Adding a data-layer to your product *In-Class Lab Tasks:* • Integrate new components into your product (e.g., payment processing, database, call to an LLM for AI features)
9	10/27	**Polishing Design** *Pre-Class Task List:* 1. Submit "Discovery Report" assignment 2. Read Course Handbook Parts 12-14 3. Read Ten Usability Hueristics (Nielsen Norman Group, 2024) — *optional* *Session Guest Workshop:* • **Jenny S. Le'Dang** (Senior Product Designer at American Airlines) *Topics Covered* • Usability and design judgment for non-designers • AI-assisted design critique and edge-case testing *In-Class Lab Tasks:* • Build a design system for your product • Conduct design iterations of your product

Session	Date	Topic
		Module 4: Pitch
10	11/3	**MVP Demo Day** *Pre-Class Task List:* 1. Read Course Handbook Part 15 2. Submit "MVP Demo" Assignment Package *In-Class Lab Tasks:* • Live presentations of MVP with peer panel feedback
11	11/10	**Developing Compelling Pitches** *Pre-Class Task List:* 1. Read Course Handbook Part 16 2. Read article: **Writing a Business Plan** (Sequoia Capital) — *optional* 3. Read article: **The 10/20/30 Rule of PowerPoint** (Guy Kawasaki, 2005) — *optional* *Topics Covered* • Anatomy of a successful pitch: narrating user value and problem solved rather than technology • Common questions investors ask startups *In-Class Lab Tasks:* • Developing your pitch • User testing of product
12	11/17	**Product Iteration & Pitch Practice** *Pre-Class Task List:* 1. Prepare preliminary version of product pitch *In-Class Lab Tasks:* • Acting on feedback (e.g., what to fix in the product, what to fix in the story) • Delivery mechanics (e.g., timing, who speaks, and answering questions under pressure)
	11/24	***Fall Break; No In-Person Class***
13	12/1	**Pitch Day**
	12/8	***No In-Person Class*** **Submit:** Final Venture Package